SPATIO-TEMPORAL MODELS WITH `INLABRU`

Pamplona (Spain) October 2026

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1 Welcome to the course!

- Welcome to the `inlabru` workshop in Pamplona!
- The aim of this workshop is to introduce you to a range of statistical modelling approaches, in particular the temporal, spatial and spatio-temporal modelling as implemented in the `inlabru` package.
- Workshop materials are available in the github repository

> ⚠ Important!
>
> Please use the instructions provided here to install and check your installation *before* the course start.

2 Learning Objectives for the workshop

At the end of the workshop, participants will have an understanding of:

- the motivation for and the challenges of analysing and modelling spatial data
- statistical models used to analyse spatial and spatio-temporal data
- the implementation of these models in the `inlabru` package
- how to independently analyse spatial data with `inlabru`

3 Intended audience

The workshop aims to cater for participants with a range of different backgrounds, who is interested in analysing data with modern spatial and spatio-temporal statistical modelling approaches.

4 Prerequisites

Participants should be familiar with the `R` environment, and general statistical approaches for modelling such as regression, analysis of (co)variance, and generalized linear models.

No knowledge of `R-INLA` or `inlabru` is required.

5 Schedule

NOTE THIS HAS TO BE UPDATED!!!!

5.0.1 Day 1

Time	Topic
10:00 - 10:30	**Welcome and intro to the course**
10:30 - 11:30	**Session 1**: Introduction to `inlabru`
11:30 - 13:00	**Session 2**: Some practical tips on `inlabru`
13:00 - 14:30	Lunch break
14:00 - 15:00	Practical Session 1
	Practical Session 2
15:00 - 16:00	**Session 3**: Time Series Models

5.0.2 Day 2

Time	Topic
8:00 - 9:00	**Session 3 cont.**: Time Series Models
9:00 - 11:00	Practical Session 3
11:00 - 12:00	**Session 4**: Introduction to Spatial Statistics
12:00 - 13:00	Lunch
13:00 - 14:00	**Session 5**: Areal processes
14.00 - 15:00	**Session 6**: Geostatistics

5.0.3 Day 3

Time	Topic
8:00 - 09:00	**Session 7**: Point Processes
09:00 - 11:00	Practical Session 4
	Practical Session 5
11:00 - 12:00	**Session 8**: Spatiotemporal models
12:00 - 13:00	Lunch break
13:00 - 14:00	**Session 9**: Model comparison and evaluation
14:00 - 15:00	Practical Session 6

5.0.4 Day 4

Time	Topic
8:00 - 9:00	**Session 10**: Multi-likelihood/joint likelihood models
9:00 - 10:00	Practical Session 7
10:00 - 11:00	**Session 11**: Zero Inflated Models
12:00 - 13:00	Lunch
13:00 - 14:00	Practical Session 7
14:00 - 15:00	Closing session and discussion

6 Licence

7 Citation

Belmont J., Martino S., Illian, J. F (2026). *inlabruCourseMarch2026: Course Materials for Noumea inlabru Course 10-14 March 2026*. https://inlabru-courses.github.io/NewCaledonia2026/

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 @Manual{Belmont2026inlabru,
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   year = {2026},
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